

Formal Verification Supplement: Quadratic Aggregation

This supplement to the paper *Quadratic Aggregation: Certificates, Finite Descriptions, and Approximation* describes five Lean packages and states how each formal theorem corresponds to a mathematical statement of the paper. Sections 1 and 2 repeat the setting and the certificate proof, so the core account is self-contained; each later section states its own system. The portable source project and the audit partitions are in `supplement/lean/`.

The packages verify: the certificate theorem under asymptotic hyperplane convexity and its HHC specialization, with the separation and cone lemmas (Section 3); closed-system properness, the whole-space Shor criterion and its consequences, exact signed-coordinate SDP tests, Dines’ theorem, and two boundary examples (Section 4); HHC of the two-ball example, its good-aggregation cone, the continuum of indispensable strict rays, and the absence of a finite weak good-aggregation description (Section 5); the exact strict and closed hulls of that example and their finite semidefinite lifts (Section 6); and its $\Theta(N^{-2})$ approximation bounds with rational meshes (Section 7).

The packages do not verify every result of the paper. Not covered are, among others, the general sharp Gram-map theorem, the obstruction for arbitrary quadratic descriptions, sufficiency of a countable dense weak family, the paper’s $2n+1$ trace-and-vector SDP test (only the signed-coordinate test is formal), the paper’s lower approximation constant $\sqrt{2}/2000$ (the formal constant is $1/2000$), the four-aggregation PDL theorem, the many-row appendix, the local-certificate application, the single-objective duality result, the sufficient criteria based on three-form convexity, the general closure equality for the Shor projection, and the paper’s other counterexamples (only the two boundary examples and the two-ball example are formal). The spectral quantitative alternative to the certificate proof is not fully formalized. Literature priority and numerical solver correctness are outside every package.

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1 Quadratic systems and the certificate question

Let $n, m \geq 1$, and let

$$f_i(x) = x^\top A_i x + 2b_i^\top x + c_i, \quad S = \{x \in \mathbb{R}^n : f_i(x) < 0 \text{ for } i = 1, \dots, m\}, \quad (1.1)$$

where $A_i \in \mathbb{S}^n$, $b_i \in \mathbb{R}^n$, and $c_i \in \mathbb{R}$. Here $\mathbb{S}^n$ denotes the real symmetric $n \times n$ matrices; $A \succeq 0$ and $A \succ 0$ mean positive semidefinite and positive definite, respectively. We use the Euclidean vector norm and the Frobenius matrix norm, and write $\langle A, B \rangle = \text{tr}(AB)$. The symbol $\text{conv } S$ denotes the ordinary convex hull, without taking its closure. A hull is *proper* if it differs from $\mathbb{R}^n$. Our question is whether a nonempty quadratic system has a proper hull, and how this can be certified by its defining inequalities.

For $\lambda \in \mathbb{R}^m$, write

$$(A_\lambda, b_\lambda, c_\lambda) = \sum_{i=1}^m \lambda_i (A_i, b_i, c_i), \quad f_\lambda = \sum_{i=1}^m \lambda_i f_i.$$

A nonzero $\lambda \geq 0$ defines an *aggregation* $f_\lambda(x) < 0$, which is valid for every $x \in S$. We call λ a *convex certificate* if $A_\lambda \succeq 0$, so that f_λ is convex on $\mathbb{R}^n$. It is *nontrivial* if $(A_\lambda, b_\lambda) \neq (0, 0)$. Otherwise the aggregated function is constant, and if $S \neq \emptyset$ evaluation at a point of S shows that the constant is strictly negative. For convenience the following cone contains zero:

$$\mathcal{K} = \{\lambda \in \mathbb{R}_+^m : A_\lambda \succeq 0\}, \quad \mathcal{T} = \{\lambda \in \mathbb{R}^m : A_\lambda = 0, b_\lambda = 0\}. \quad (1.2)$$

Thus a nontrivial convex certificate exists precisely when $\mathcal{K} \not\subseteq \mathcal{T}$.

The homogeneous matrices and map associated with (1.1) are

$$Q_i = \begin{pmatrix} A_i & b_i \\ b_i^\top & c_i \end{pmatrix}, \quad f_i^h(x, t) = x^\top A_i x + 2tb_i^\top x + c_i t^2, \quad f^h = (f_1^h, \dots, f_m^h). \quad (1.3)$$

We also write $Q_\lambda = \sum_i \lambda_i Q_i$ and $g_i(x) = x^\top A_i x$. Define the strict homogeneous feasible set $S^h = \{(x, t) : f_i^h(x, t) < 0 \text{ for all } i\}$. It is open and invariant under multiplication by every nonzero real scalar. For $t \neq 0$, $(x, t) \in S^h$ is equivalent to $x/t \in S$. For $\alpha \in \mathbb{R}^n \setminus \{0\}$ and $s \in \mathbb{R}$, put

$$H_{\alpha, s} = \{(x, t) \in \mathbb{R}^{n+1} : \alpha^\top x = st\}, \quad E = \{(x, t) : t = 0\}. \quad (1.4)$$

Definition 1.1 (Hidden hyperplane convexity). A homogeneous quadratic map $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^m$ has *hidden hyperplane convexity* (HHC) if $\phi(H)$ is convex for every linear hyperplane $H \subset \mathbb{R}^d$.

This is the definition of Blekherman, Dey, and Sun [3, Definition 2.1], published as [4]. Ordinary *hidden convexity* means only that $\phi(\mathbb{R}^d)$ is convex; the two properties differ. Every homogeneous quadratic image contains zero and is invariant under nonnegative scaling, so a convex such image is a convex cone, not necessarily closed. HHC implies ordinary hidden convexity when $d \geq 3$: any two points of $\mathbb{R}^d$ lie in some linear hyperplane, whose image contains the segment joining their images.

Question 1.2 (Blekherman–Dey–Sun, Conjecture 3.3). Suppose $S \neq \emptyset$ and f^h has HHC. Is

$$\text{conv } S \neq \mathbb{R}^n \iff \exists \lambda \in \mathbb{R}_+^m : A_\lambda \succeq 0, \quad (A_\lambda, b_\lambda) \neq (0, 0)? \quad (1.5)$$

The formulation in Blekherman et al. [3, Conjecture 3.3] says that the aggregated function is not a negative constant; under nonemptiness this is equivalent to the displayed condition. Their standing setting has $n \geq 3$ and $m \geq 2$; ours includes all $n, m \geq 1$. Numbered locators for that work refer to the May 29, 2023 arXiv version; the earlier author manuscript numbers several background results differently.

The reverse implication in (1.5) is elementary: a nonconstant convex quadratic aggregation bounds S by a proper convex sublevel set. Under convexity of $(g_1, \dots, g_m)(\mathbb{R}^n)$, Blekherman, Dey, and Sun prove that $\mathcal{K} = \{0\}$ implies $\text{conv } S = \mathbb{R}^n$ [3, Proposition 2.14], and for $n \geq 2$ their separable case, with all Q_i diagonal, gives the certificate characterization [3, Theorem 2.12]. The unresolved case is the forward implication when $\mathcal{K}$ contains nonzero vectors but all of them belong to $\mathcal{T}$.

The certificate question differs from a complete description of the hull. In the aggregation literature, a *good aggregation* is a nonnegative nonzero multiplier λ such that $\text{conv } S \subseteq \{x : f_\lambda(x) < 0\}$ and Q_λ has at most one negative eigenvalue. The function f_λ need not be globally convex. Blekherman, Dey, and Sun prove that these good aggregations describe $\text{conv } S$ under HHC, $n \geq 3$, and $\emptyset \neq S$ with proper hull [3, Theorem 2.9].

The two-constraint case is classical. Dines' theorem gives convexity of the joint image of two homogeneous quadratic forms [6], and therefore HHC for any pair of homogeneous forms and their restrictions. Yildiran describes the hull of a nonempty set defined by two strict quadratic inequalities using at most two aggregations [7]. For three quadratic inequalities in dimension $n \geq 3$ with a nonempty proper hull, Dey, Muñoz, and Serrano obtain an aggregation description when Q_1, Q_2, Q_3 admit a positive definite real linear combination, with coefficients of either sign [5]. Blekherman and Dunbar use the spectral curve and algebraic topology to obtain further three-quadratic hull descriptions and finiteness results [1, 2]. Neither Dines' theorem nor the elementary separation and compactness facts used below are contributions of this paper.

2 The certificate theorem

To answer Question 1.2, only a family of hyperplanes approaching E is needed. The following weaker assumption makes this precise.

Definition 2.1 (Asymptotic hyperplane convexity). The map f^h has *asymptotic hyperplane convexity* (AHC) if, for every $\alpha \neq 0$ and every $R \in \mathbb{R}$, there exists $s > R$ such that $f^h(H_{\alpha,s})$ is convex.

Equivalently, for each $\alpha \neq 0$ there is a strictly increasing sequence (s_k) tending to $+\infty$ with all these images convex. Indeed, the sequence condition implies the unbounded-level condition; conversely, choose a good $s_1 > 1$ and recursively a good $s_{k+1} > \max\{s_k, k+1\}$. HHC implies AHC because $H_{\alpha,s}$ is a linear hyperplane for every s . This assumption concerns the images themselves and requires no closedness.

Theorem 2.2 (Convex aggregation certificate). *Suppose $S \neq \emptyset$ and f^h has AHC. Then*

$$\text{conv } S \neq \mathbb{R}^n \iff \exists \lambda \geq 0 : \quad A_\lambda \succeq 0, \quad (A_\lambda, b_\lambda) \neq (0, 0). \quad (2.1)$$

The certificate-to-proper-hull implication holds without either assumption.

Corollary 2.3 (Conjecture 3.3 of Blekherman–Dey–Sun). *If $S \neq \emptyset$ and f^h has HHC, then (2.1) holds.*

The corollary follows because HHC implies AHC. It answers the cited conjecture for every finite number of inequalities, including the case where nonzero convex certificates exist but all of them have vanishing quadratic and linear coefficients; the proof shows that this case forces a whole-space hull.

2.1 Hyperplanes beyond a supporting halfspace

Lemma 2.4 (No common strict negative quadratic direction). *If $\text{conv } S \neq \mathbb{R}^n$, then there is no $v \in \mathbb{R}^n$ satisfying $v^\top A_i v < 0$ for every i . Equivalently, $E \cap S^h = \emptyset$.*

Proof. If such a v existed, then for any fixed $y \in \mathbb{R}^n$,

$$f_i(y + rv) = r^2 v^\top A_i v + 2r(v^\top A_i y + b_i^\top v) + f_i(y) \longrightarrow -\infty \quad (|r| \longrightarrow \infty).$$

There are finitely many constraints. Hence both $y + rv$ and $y - rv$ belong to S for all sufficiently large positive r , and their midpoint y belongs to $\text{conv } S$. This holds for every y , a contradiction. Finally, $f_i^h(v, 0) = v^\top A_i v$ proves the equivalence. $\square$

The lemma concerns a simultaneous strict sign condition on the leading forms; it does not characterize the recession cone of S .

Lemma 2.5 (Sweeping hyperplanes). *Suppose $S \neq \emptyset$ and $\text{conv } S \neq \mathbb{R}^n$. There are $\alpha \neq 0$ and $\beta \in \mathbb{R}$ such that*

$$\alpha^\top x < \beta \quad (x \in S). \quad (2.2)$$

For any such α, β , one has $H_{\alpha, s} \cap S^h = \emptyset$ for every $s \geq \beta$.

Proof. The set S is open, and its convex hull is open as well: in a finite convex combination, one point with positive weight can vary in an open ball, giving an open ball around the combination. Separate a point $y \notin \text{conv } S$ from this nonempty open convex set. Finite-dimensional separation supplies $\alpha \neq 0$ with $\alpha^\top x \leq \alpha^\top y$ throughout $\text{conv } S$. Equality at a point of that open set would be contradicted by a small move in direction α . Thus (2.2) holds with $\beta = \alpha^\top y$.

If $(x, t) \in H_{\alpha, s} \cap S^h$ and $t = 0$, then Lemma 2.4 gives a contradiction. If $t \neq 0$, the identity $f_i^h(x, t) = t^2 f_i(x/t)$ gives $x/t \in S$, whereas $\alpha^\top(x/t) = s \geq \beta$, again a contradiction. This reasoning applies to either sign of t . $\square$

Lemma 2.6 (Separation on a hyperplane). *Suppose $s \neq 0$, the set $f^h(H_{\alpha, s})$ is convex, and $H_{\alpha, s} \cap S^h = \emptyset$. There exists $\lambda \geq 0$ with $\sum_i \lambda_i = 1$ such that, for every $x \in \mathbb{R}^n$,*

$$0 \leq x^\top A_\lambda x + \frac{2}{s}(\alpha^\top x)(b_\lambda^\top x) + \frac{c_\lambda}{s^2}(\alpha^\top x)^2. \quad (2.3)$$

Proof. The convex image $C = f^h(H_{\alpha, s})$ contains zero and is disjoint from the open negative orthant $N = -\mathbb{R}_{++}^m$. Separation of these two convex sets gives a nonzero λ and a real a with $\lambda^\top z \leq a \leq \lambda^\top y$ for $z \in N$, $y \in C$. If some $\lambda_i < 0$, sending z_i to $-\infty$ with the other coordinates fixed contradicts the upper bound. Thus $\lambda \geq 0$, and $\sup_{z \in N} \lambda^\top z = 0$. Since $0 \in C$, necessarily $a = 0$, and $\lambda^\top f^h \geq 0$ on $H_{\alpha, s}$. Divide λ by its positive coordinate sum. Substituting $(x, t) = (x, \alpha^\top x/s)$ gives (2.3). Closedness of C was not used. $\square$

2.2 A nonzero limit in coefficient space

The relevant cone retains the quadratic and linear coefficients together:

$$P = \text{cone}\{(A_i, b_i) : 1 \leq i \leq m\} = \{(A_\lambda, b_\lambda) : \lambda \geq 0\} \subseteq \mathbb{S}^n \times \mathbb{R}^n. \quad (2.4)$$

We equip this space with $\|(A, b)\| = (\|A\|_F^2 + \|b\|^2)^{1/2}$. Closedness of this cone is essential, because a limit must correspond to actual aggregation weights. The following elementary fact supplies it.

Lemma 2.7 (Finite coefficient cones are closed). *The cone generated by a finite family in a finite-dimensional real vector space is closed.*

Proof. Every conic combination can be represented using a linearly independent subfamily. To see this, choose a representation with the fewest positive coefficients. If its active generators are dependent, take a nonzero relation $\sum_i d_i v_i = 0$, reversing its sign so some $d_i > 0$. For the active coefficients $a_i > 0$, set $\tau = \min_{d_i > 0} a_i/d_i$. Then all $a_i - \tau d_i$ are nonnegative, at least one vanishes, and the represented point is unchanged, a contradiction. For a linearly independent subfamily, the coefficient map is a linear homeomorphism onto its span; that span is closed, so the image of the nonnegative orthant is closed. There are only finitely many subfamilies. The original cone is their finite union and is closed. The empty subfamily accounts for the zero vector. $\square$

Proof of Theorem 2.2. First suppose a nontrivial convex certificate λ exists. Its nonconstant coefficient pair implies $\lambda \neq 0$, so $f_\lambda < 0$ on S . Positive semidefiniteness makes f_λ convex, whence $\text{conv } S \subseteq \{f_\lambda < 0\}$. This sublevel set is proper. Indeed, if $A_\lambda \neq 0$, it has a unit eigenvector v with eigenvalue $\mu > 0$, and $f_\lambda(rv) = \mu r^2 + 2rb_\lambda^\top v + c_\lambda \rightarrow +\infty$ as $r \rightarrow +\infty$. If $A_\lambda = 0$, then $b_\lambda \neq 0$ and $f_\lambda(rb_\lambda) = 2r\|b_\lambda\|^2 + c_\lambda \rightarrow +\infty$. This proves the easy implication without using nonemptiness or AHC.

For the converse, suppose $S \neq \emptyset$, f^h has AHC, and $\text{conv } S \neq \mathbb{R}^n$. Fix $x_0 \in S$, and choose α, β from Lemma 2.5. AHC gives positive levels $s_k \rightarrow +\infty$ with $s_k \geq \beta$ and convex images $f^h(H_{\alpha, s_k})$. Lemmas 2.5 and 2.6 give nonzero normalized weights $\lambda^k \geq 0$ satisfying (2.3). Write $(A_k, b_k, c_k) = (A_{\lambda^k}, b_{\lambda^k}, c_{\lambda^k})$. Strict feasibility gives the crucial upper bound

$$c_k < -x_0^\top A_k x_0 - 2b_k^\top x_0 =: L(A_k, b_k), \quad (2.5)$$

where L is a fixed continuous linear functional on coefficient space. The multiplier of c_k in (2.3) is a square. Replacing c_k by its upper bound therefore yields

$$0 \leq x^\top A_k x + \frac{2}{s_k}(\alpha^\top x)(b_k^\top x) + \frac{L(A_k, b_k)}{s_k^2}(\alpha^\top x)^2 \quad (x \in \mathbb{R}^n). \quad (2.6)$$

Moreover, $(A_k, b_k) \neq (0, 0)$: otherwise (2.5) gives $c_k < 0$, and the original inequality (2.3) at $x = \alpha$ reads $0 \leq c_k \|\alpha\|^4 / s_k^2 < 0$.

Set $\rho_k = \|(A_k, b_k)\| > 0$ and $(\hat{A}_k, \hat{b}_k) = (A_k, b_k)/\rho_k$. These pairs lie in the unit section of P , which is compact by Lemma 2.7. Passing to a subsequence, let

$$(\hat{A}_k, \hat{b}_k) \longrightarrow (A_*, b_*) \in P, \quad \|(A_*, b_*)\| = 1.$$

Divide (2.6) by ρ_k . By linearity of L , its right-hand side becomes

$$x^\top \hat{A}_k x + \frac{2}{s_k}(\alpha^\top x)(\hat{b}_k^\top x) + \frac{L(\hat{A}_k, \hat{b}_k)}{s_k^2}(\alpha^\top x)^2.$$

For every fixed x the last two terms tend to zero, so $x^\top A_* x \geq 0$ for every x ; the same coefficient subsequence serves every x . Hence the symmetric matrix A_* is positive semidefinite. Since $(A_*, b_*) \in P$, there are actual weights $\lambda^* \geq 0$ with $(A_{\lambda^*}, b_{\lambda^*}) = (A_*, b_*)$. Their coefficient pair has norm one, so they give the required nontrivial convex certificate. $\square$

Remark 2.8 (What the limit proves). Normalizing the weights in the simplex alone can give a negative constant aggregation in the limit. Normalizing the nonconstant coefficient pair prevents this loss of information. Bound (2.5) removes the constant before normalization; no bound on c_k/ρ_k is assumed. The final weights need not be a limit of the original weights, or of their rescalings. Their existence follows from closedness of the finitely generated cone P .

Remark 2.9 (Convexity along one normal). The hard implication needs convexity of $f^h(H_{\alpha,s})$ at arbitrarily large levels for only one α satisfying (2.2). AHC is a convenient condition that can be stated without knowing a supporting normal. For fixed α , the hyperplanes $H_{\alpha,s}$ approach E as $s \rightarrow +\infty$; convexity of $f^h(E)$ alone is not the hypothesis used in the proof. There is no restriction beyond $n, m \geq 1$. For one constraint, a negative quadratic direction gives a whole-space hull by Lemma 2.4; the remaining cases follow from the easy implication or a negative constant. For two constraints, Dines' theorem makes HHC automatic, recovering the known certificate case discussed in Section 1.

2.3 A quantitative form of the obstruction

We give a second way to exclude degeneration to constant certificates. If all convex certificates are trivial, it bounds the sweep level uniformly. The cone estimate is a standard compactness argument.

Lemma 2.10 (Uniform cone separation). *Let P, D be closed cones containing zero in a finite-dimensional normed space, with $P \cap D = \{0\}$. Then some $\kappa > 0$ satisfies*

$$\text{dist}(p, D) \geq \kappa \|p\| \quad (p \in P).$$

Neither cone is required to be convex.

Proof. If $P = \{0\}$, any positive κ works. Otherwise the continuous function $p \mapsto \text{dist}(p, D)$ has a positive minimum κ on the compact set $P \cap \{\|p\| = 1\}$: a zero distance would put a unit point of P in the closed set D . Scaling both cones proves the bound for $p \neq 0$, and the assertion at zero follows from $0 \in D$. $\square$

Apply the lemma to P in (2.4) and $D = \mathbb{S}_+^n \times \mathbb{R}^n$. The condition $P \cap D = \{0\}$ is exactly the assertion that every convex certificate is trivial. If A_- is the positive semidefinite negative part of a symmetric matrix A , then

$$\text{dist}((A, b), D) = \|A_-\|_F \leq \sqrt{n} \max\{0, -\lambda_{\min}(A)\}. \quad (2.7)$$

For the equality, diagonalize A orthogonally. In that basis every positive semidefinite approximant has nonnegative diagonal entries; the sum of squared errors is at least the sum of squares of the negative eigenvalues of A . Replacing those eigenvalues by zero achieves this bound, and the b component can be left unchanged. The inequality follows by bounding each negative eigenvalue in absolute value by $\max\{0, -\lambda_{\min}(A)\}$.

Proposition 2.11 (Bound on a sweep certificate). *Suppose every convex certificate is trivial, and let κ be as in Lemma 2.10 for the preceding cones. If $\lambda \geq 0$, $c_\lambda < 0$, $s > 0$, $\alpha \neq 0$, and (2.3) holds, then*

$$s \leq \frac{2\sqrt{n}\|\alpha\|}{\kappa}. \quad (2.8)$$

Proof. The inequality at $x = \alpha$ shows that $\rho = \|(A_\lambda, b_\lambda)\|$ cannot vanish. The cone estimate and (2.7) give $\lambda_{\min}(A_\lambda) \leq -\kappa\rho/\sqrt{n}$. Choose a unit eigenvector u for this eigenvalue. Evaluating (2.3) at u , discarding the nonpositive constant term, and using $\|b_\lambda\| \leq \rho$ gives

$$0 \leq -\frac{\kappa}{\sqrt{n}}\rho + \frac{2\|\alpha\|}{s}\rho.$$

Division by $\rho > 0$ proves (2.8). □

This gives an alternative completion of the hard implication. Under its hypotheses, suppose every convex certificate is trivial and choose the normalized sweep weights λ^k as before. Compactness of the simplex gives a subsequence converging to $\bar{\lambda}$, with $\bar{\lambda} \geq 0$ and $\sum_i \bar{\lambda}_i = 1$. The coefficients b_k, c_k are bounded, so letting $k \rightarrow \infty$ in (2.3) shows $A_{\bar{\lambda}} \succeq 0$. Triviality then gives $A_{\bar{\lambda}} = 0$, $b_{\bar{\lambda}} = 0$, and $c_{\bar{\lambda}} < 0$ by strict feasibility. Thus $c_k < 0$ eventually along this subsequence. Proposition 2.11 contradicts $s_k \rightarrow +\infty$. The constant κ depends only on the original quadratic and linear coefficients; this argument neither bounds aggregation weights nor provides a procedure for verifying AHC.

3 Formal verification of the certificate theorem

A Lean development verifies Theorem 2.2 and its HHC specialization, Corollary 2.3, for finite families of real symmetric matrices, real vectors, and real constants. Its strict feasible set, ordinary convex hull, and nontrivial certificate agree with those defined here. The main declaration is `proper_hull_iff_certificate` in the namespace `QuadraticAggregation.System`; the HHC specialization is `proper_hull_iff_certificate_of_hhc`. Their assumptions are nonemptiness and AHC, or nonemptiness and HHC, respectively; no positive definite combination is assumed. The certificate-to-proper-hull implication is proved separately without either assumption.

The formal definitions express HHC through kernels of nonzero linear functionals, which are precisely the linear hyperplanes in this finite-dimensional setting. AHC is stated using arbitrarily large good levels, as in Definition 2.1; its equivalence with the increasing sequence formulation is also proved. Matrix positive semidefiniteness is linked to nonnegativity of the quadratic form, using the symmetry of the original matrices and their aggregations.

The proof follows the constant-elimination and coefficient-normalization argument of Section 2. It constructs the separation weights from the convex hyperplane images and obtains the nonzero limit in the actual finitely generated coefficient cone; these are proved steps, not additional hypotheses. The implementation uses the entrywise maximum matrix norm and the product maximum norm for coefficient pairs. These are equivalent to the Frobenius and Euclidean norms in finite dimension; the compactness proof works in either choice.

The verified supporting results include Lemmas 2.4, 2.5, 2.6, and 2.7, and the uniform cone-separation lemma, Lemma 2.10. The last lemma is checked as a separate result, although the main formal proof does not use it. The spectral distance calculation and the quantitative alternative in Section 2.3 are not formalized as separate results.

This core package consists of eleven modules in `supplement/lean/Formal/QuadraticAggregation/`, with a declaration coverage map and reproducible checks. With the pinned Lean 4.33.1 toolchain, warning-free targeted compilation, a transitive axiom audit of all 178 declarations owned by those modules, and kernel replay of each module passed; these checks were rerun for this manuscript with unchanged source fingerprints. The only permitted axioms were `propext`, `Classical.choice`, and `Quot.sound`. Kernel replay uses the pinned Lean kernel and imported dependencies; it is neither verification by an independently implemented proof assistant nor a replay of all dependencies from source.

This package concerns existence of a nontrivial globally convex aggregation, not a description of the entire hull by such aggregations. The closed-system and relaxation consequences, SDP characterizations, examples, applications, further aggregation results, and literature-priority claims are outside it; separate packages, described in the accompanying formal accounts, verify specified later results.

4 Formal verification of closed-system and semidefinite consequences

The second Lean package verifies consequences of the certificate theorem and two examples that delimit them. It uses the actual closed set $T = \{x : f_i(x) \leq 0 \text{ for every } i\}$ and the ordinary convex hull. Its Shor projection is

$$P = \left\{ x : \exists X, \begin{pmatrix} 1 & x^\top \\ x & X \end{pmatrix} \succeq 0, \quad \text{tr}(A_i X) + 2b_i^\top x + c_i \leq 0 \text{ for every } i \right\}.$$

The formalization proves the Schur-complement equivalence with covariance slack $Y \succeq 0$, where $X = xx^\top + Y$ and the lifted inequalities are $f_i(x) + \text{tr}(A_i Y) \leq 0$. It also proves $S \subseteq T \subseteq P$ and convexity of P .

Consequences and their assumptions. If $S \neq \emptyset$ and asymptotic hyperplane convexity holds, then

$$\text{conv } T \neq \mathbb{R}^n \iff \exists \lambda \geq 0, \lambda \neq 0 : A_\lambda \succeq 0, (A_\lambda, b_\lambda) \neq (0, 0),$$

and

$$P = \mathbb{R}^n \iff \text{conv } S = \mathbb{R}^n \iff \text{conv } T = \mathbb{R}^n.$$

Here asymptotic hyperplane convexity means that, for every nonzero normal, the homogeneous image is convex on arbitrarily distant sweeping hyperplanes; HHC implies this condition. A PSD aggregate is nonpositive throughout P , because the trace pairing of two PSD matrices is nonnegative. A nonconstant PSD quadratic is unbounded above, so its nonpositive sublevel set is proper. These facts prove the certificate-to-properness implications without any feasibility or hyperplane-convexity premise.

An unconditional Shor characterization. With strict feasibility alone, the package proves

$$P = \mathbb{R}^n \iff \text{every nonnegative PSD aggregation has } A_\lambda = b_\lambda = 0.$$

The verified converse uses open-set separation and does not require a closed linear image of the PSD cone. For fixed x , suppose there were no $Y \succeq 0$ satisfying all $f_i(x) + \text{tr}(A_i Y) < 0$. Separate the convex affine image of PSD matrices from the open negative orthant. This gives nonzero $\lambda \geq 0$ with

$$0 \leq f_\lambda(x) + \text{tr}(A_\lambda Y) \quad (Y \succeq 0).$$

Scaling the rank-one matrices $Y = vv^\top$ proves $A_\lambda \succeq 0$, and setting $Y = 0$ gives $f_\lambda(x) \geq 0$. If every such aggregation is trivial, evaluation at a strict feasible point instead gives $f_\lambda(x) = c_\lambda < 0$, a contradiction. Thus every fiber admits an actual PSD slack with strict inequalities, which proves membership in the ordinary Shor projection, not only in its closure.

Exact finite SDP characterization. Normalize multipliers by $\lambda \geq 0$ and $\sum_i \lambda_i = 1$, and require $A_\lambda \succeq 0$. The resulting set is compact or empty. There are $n(n+1)/2 + n$ independent coordinates of (A_λ, b_λ) . The formalization proves that every signed coordinate objective attains its maximum on a nonempty normalized set. All these coordinates vanish throughout that set exactly when every one of the $n(n+1) + 2n$ signed programs has optimum zero, or the common feasible set is empty. The normalization argument connects this to the full multiplier cone and the preceding whole-space characterizations. These are mathematical equivalences about exact feasibility and exact optimum values. They do not verify floating-point SDP solvers or turn a near-zero objective into a zero certificate.

Verified boundary examples. The first example uses

$$f_1 = x_1 x_2 - 1, \quad f_2 = 1 - x_1 x_2, \quad f_3 = x_1^2 - x_2^2 \quad \text{on } \mathbb{R}^3.$$

Its strict system is empty, its closed set is nonempty, and its closed feasible hull lies in $|x_1| \leq 1$. Every PSD aggregation has zero quadratic and linear parts. Nevertheless HHC holds. Thus strict feasibility cannot be removed from the closed-system consequence.

The second example uses $f_1 = x_1^2 - 4$ and $f_2 = -x_1^2 + 2x_1 + 3$ on $\mathbb{R}^3$. Its strict feasible set is the convex strip $-2 < x_1 < -1$, and HHC holds. Every nonzero nonnegative globally convex aggregation is strictly negative at zero. Zero also belongs to the Shor projection: $X = \text{diag}(7/2, 0, 0)$ gives a PSD block and both lifted residuals equal $-1/2$. Hence that projection is proper but strictly larger than the hull. The examples include proofs of HHC: a formal algebraic proof of Dines' two-form theorem establishes HHC for pairs of forms, and a linear-image argument handles the repeated negative form.

Scope. The proof files extend `Formal.QuadraticAggregation`. The frozen claims, declaration coverage, and audit inputs are in `supplement/lean/audits/28-quadratic-aggregation-consequences/`. The full good-aggregation hull theorem, the general sufficient criteria based on classical three-form convexity, additional examples, numerical solver correctness, and literature priority are outside this package. The infinite-aggregation construction has a separate verification scope.

5 Formal verification of the infinite-aggregation example

The third Lean package concerns the explicit family

$$S_r = \{(u, v) \in \mathbb{R}^r \times \mathbb{R}^r : \|u\|^2 < 1, \|v\|^2 < 1, u^\top v > 1/2\}, \quad r \geq 2.$$

Squared lengths and inner products in the formalization are Euclidean coordinate sums. The package treats the actual quadratic map and ordinary convex hull, including the four-variable case $r = 2$.

Geometry and good multipliers. The set S_r is nonempty and bounded, and its ordinary convex hull is proper. Its homogeneous map is

$$(u, v, t) \mapsto (\|u\|^2 - t^2, \|v\|^2 - t^2, t^2/2 - u^\top v).$$

The formal HHC assertion requires convexity of this image on the kernel of every nonzero real linear functional. It is proved for all $r \geq 2$; HHC is not supplied as an additional premise for this example.

The proof uses an explicit orthonormal frame and an algebraic unit-circle construction to obtain the full range of a linear functional on a fixed two-column Gram fiber. For a hyperplane $\text{tr}(P^\top U) + st = 0$, put $G = U^\top U$, $H = P^\top P$, and $T = t^2$. Its exact feasible Gram domain is

$$G \succeq 0, \quad T \geq 0, \quad s^2 T \leq \text{tr}(HG) + 2\sqrt{\det H \det G}.$$

Both necessity and realization are proved, including singular Gram matrices and zero coefficients. Concavity of the square root of the two-by-two determinant proves convexity of this domain. The homogeneous quadratic image is its linear image, which gives HHC.

A good multiplier is defined by the original conditions: it is nonzero and nonnegative, its homogeneous aggregate has at most one negative eigenvalue counted with multiplicity, and its strict aggregation contains $\text{conv } S_r$. The proved classification is

$$\lambda \text{ is good} \iff \lambda \geq 0, \quad \lambda \neq 0, \quad \lambda_3^2 \leq 4\lambda_1\lambda_2.$$

The homogeneous matrix consists of r copies of

$$B_\lambda = \begin{pmatrix} \lambda_1 & -\lambda_3/2 \\ -\lambda_3/2 & \lambda_2 \end{pmatrix}$$

and the scalar $\lambda_3/2 - \lambda_1 - \lambda_2$, up to coordinate order. For nonnegative multipliers violating the discriminant inequality, the repeated block gives at least r negative eigenvalues. For nonzero nonnegative multipliers satisfying it, the block is PSD and the scalar is strictly negative, so there is exactly one negative eigenvalue. Convexity of the original aggregate and its strict negativity on S_r establish strict negativity throughout the ordinary convex hull. Thus the classification does not assume a general aggregation hull-description theorem.

Indispensable strict rays. For every $\tau \in [1, 2]$, the package constructs actual vectors whose Gram matrix is

$$\begin{pmatrix} 1 - 1/(10\tau) & 2/5 \\ 2/5 & 1 - \tau/10 \end{pmatrix}.$$

Their point x_τ has residual vector $\frac{1}{10}(-1/\tau, -\tau, 1)$. Every good aggregate is nonpositive there, with equality exactly for positive multiples of $\rho(\tau) = (\tau, 1/\tau, 2)$. The multiplier $\rho(\tau)$ is itself good, so $x_\tau \notin \text{conv } S_r$; every good multiplier on a different ray is strictly satisfied at this point.

Consequently, if a family of strict good aggregations describes $\text{conv } S_r$ exactly, it contains a positive multiple of $\rho(\tau)$ for every $\tau \in [1, 2]$. The formal cardinality argument normalizes weights to third coordinate two and proves that the set of normalized weights is uncountable. This counts distinct rays, not repeated indices in a family. In particular, no countable family of strict good aggregations describes the ordinary hull. This statement is conditional on an exact description; the package does not separately prove that the intersection of all good strict aggregations equals the hull.

The closed hull requires a different witness. For any finite family of good multipliers, choose an omitted ray and decrease the Gram off-diagonal $2/5$ by a sufficiently small positive amount. The perturbed Gram matrix remains positive definite and is realized by actual vectors. Every selected aggregate is strictly negative there, while the omitted aggregate is strictly positive. Every good aggregate is nonpositive on $\text{cl}(\text{conv } S_r)$ by continuity. Hence the perturbed point is outside this closed hull, although it satisfies every selected weak aggregation. No finite intersection of weak good aggregations can therefore equal $\text{cl}(\text{conv } S_r)$.

Scope and records. The Lean sources are in `Formal.InfiniteAggregation`; the frozen claims and declaration coverage are in `supplement/lean/audits/29-infinite-aggregation/`. This package does not establish an exact hull formula, countable weak aggregation sufficiency, equality with the hull of the original weak system, a finite SDP lift, an obstruction to arbitrary quadratic descriptions, approximation bounds, or the general sharp Gram-map theorem. It also makes no literature-priority claim.

6 Verified exact hulls and finite semidefinite lifts

For $r \geq 2$, let

$$S = \{(u, v) \in \mathbb{R}^r \times \mathbb{R}^r : \|u\|^2 < 1, \|v\|^2 < 1, u^\top v > 1/2\},$$

and let T denote the system obtained by replacing all three inequalities by weak inequalities. Write $p = 1 - \|u\|^2$, $q = 1 - \|v\|^2$, and $d = u^\top v$. This Lean package proves

$$\begin{aligned} \text{conv } S &= \{(u, v) : p > 0, q > 0, d + \sqrt{pq} > 1/2\}, \\ \text{cl}(\text{conv } S) &= \text{conv } T = \{(u, v) : p \geq 0, q \geq 0, d + \sqrt{pq} \geq 1/2\}. \end{aligned}$$

Every equality includes $r = 2$. The proof of sufficiency is a direct two-point construction, so it does not depend on an unformalized general good-aggregation hull theorem.

A two-point proof in every required dimension. Suppose the strict scalar conditions hold. Put $a = \sqrt{p}$ and $b = \sqrt{q}$. Choose a unit vector e perpendicular to $bu - av$; this requires only $r \geq 2$. Orthogonality implies

$$\frac{u^\top e}{a} = \frac{v^\top e}{b} = h.$$

Choose

$$\max \left\{ 0, \frac{1/2 - d}{ab} \right\} < k < 1, \quad t_\pm = -h \pm \sqrt{h^2 + k}.$$

The two roots have opposite signs and satisfy $2ht_\pm + t_\pm^2 = k$. For $x(t) = (u + tae, v + tbe)$ and $t \in \{t_-, t_+\}$, direct expansion gives

$$\|u + tae\|^2 = \|u\|^2 + pk, \quad \|v + tbe\|^2 = \|v\|^2 + qk, \quad (u + tae)^\top (v + tbe) = d + abk.$$

Thus both $x(t_-)$ and $x(t_+)$ lie in S . The positive weights $t_+/(t_+ - t_-)$ and $-t_-/(t_+ - t_-)$ sum to one and recover (u, v) . This proves hull sufficiency, including the four-variable case.

Necessity follows from the already verified validity of every good-cone aggregation on $\text{conv } S$. The coordinate multipliers give $p, q > 0$; the multiplier $(q, p, 2\sqrt{pq})$ gives $d + \sqrt{pq} > 1/2$. The package also proves the weak cone-test implication, with explicit separating multipliers when $p = 0$ or $q = 0$. It does not infer the weak claim by replacing strict signs in the strict proof.

Actual affine matrix lifts. Define the symmetric matrix

$$L(u, v, \sigma) = \begin{pmatrix} 1 & \sigma & u^\top \\ \sigma & 1 & v^\top \\ u & v & I_r \end{pmatrix}.$$

The formalization proves both the PSD and positive definite Schur-complement equivalences for this matrix and eliminates σ . Consequently,

$$\begin{aligned}(u, v) \in \text{conv } S &\iff \exists \sigma > 1/2 : L(u, v, \sigma) \succ 0, \\ (u, v) \in \text{cl}(\text{conv } S) &\iff \exists \sigma \geq 1/2 : L(u, v, \sigma) \succeq 0.\end{aligned}$$

The latter representation also describes $\text{conv } T$. Singular Schur complements and both signs of $\sigma - d$ are included.

Closure and the original weak system. For a point in the weak scalar region and $0 \leq s < 1$, the scalar margin of (su, sv) above $1/2$ is at least $(1 - s^2)/2$. This follows from concavity of the square root of the two-by-two determinant. Thus radial contraction approximates every weak-region point by strict-hull points. Continuity gives the reverse closure inclusion.

To prove equality with $\text{conv } T$, the proof uses compactness of T and of the image of $[0, 1] \times T \times T$ under $(\alpha, y, z) \mapsto \alpha y + (1 - \alpha)z$. The two-point construction places the strict hull inside this compact image, hence its closure lies there as well. The image is contained in $\text{conv } T$; convexity of the weak scalar region gives the other inclusion.

Finally, the package identifies $\text{conv } S$ with the intersection of all strict good aggregations, and $\text{cl}(\text{conv } S)$ with the intersection of all weak good aggregations. Goodness retains its source definition using the actual homogeneous eigenvalue count and ordinary-hull validity. These equalities supply the exact descriptions for which the preceding package proved that no countable strict good description and no finite weak good description can suffice.

Scope. The frozen claims and declaration coverage are in `supplement/lean/audits/30-infinite-aggregation-hull/`.

These sources extend `Formal.InfiniteAggregation` without changing the earlier certified modules. This package does not verify a numerical SDP solver, quantitative approximation, arbitrary-quadratic impossibility, or a general aggregation hull theorem. The accuracy bounds have a separate verification package.

7 Verified accuracy of finite good aggregations

For $u, v \in \mathbb{R}^r$, $r \geq 2$, define

$$f(u, v) = (\|u\|^2 - 1, \|v\|^2 - 1, 1/2 - u^\top v), \quad S = \{x : f_i(x) < 0 \text{ for all } i\}, \quad C = \text{cl}(\text{conv } S).$$

The preceding verified classification identifies good multipliers exactly as nonzero $\lambda \geq 0$ with $\lambda_3^2 \leq 4\lambda_1\lambda_2$. Write $f_\lambda = \sum_i \lambda_i f_i$. For a finite set F of good multipliers, put

$$P_F = \{x : f_\lambda(x) \leq 0 \text{ for every } \lambda \in F\}, \quad e_N = \inf_{|F| \leq N} d_H(C, P_F).$$

Here d_H is the extended Hausdorff distance in the Euclidean norm on all $2r$ original coordinates. It is infinite for an unbounded relaxation. The infimum ranges over arbitrary good multipliers, including interior ones and arbitrary positive rescalings; no minimizing family is assumed.

The Lean package proves, for every $N \geq 2$,

$$\frac{\sqrt{2}(\log 2)^2}{1600N^2} \leq e_N \leq \frac{5\sqrt{2}\pi^2}{16(N-1)^2}.$$

The constants are independent of r . A linear homeomorphism into an explicit Euclidean space proves agreement with the original variables, their quadratic residuals, and their actual closed convex hull. The default maximum norm on a function space is not used in these error bounds.

A uniform upper bound. The angle multipliers

$$\lambda(\theta) = (\cos^2 \theta, \sin^2 \theta, 2 \sin \theta \cos \theta), \quad 0 \leq \theta \leq \pi/2,$$

are good, and all their weak inequalities describe C . The formal proof includes the coordinate endpoints and does not require the associated two-by-two slack matrix to be PSD: only its nonnegative directions are tested.

For a sample set with covering radius ρ , the angular quadratic $h_x(\theta) = -f_{\lambda(\theta)}(x)$ is bounded below by $-5\rho^2$ throughout the interval whenever the sampled cuts and both coordinate bounds hold. The proof minimizes this continuous function. At an interior minimum, its derivative is zero; an exact rotation identity and $|\sin t| \leq |t|$ give the bound. Endpoint minima are already nonnegative. This is an elementary alternative to the paper's second-derivative Taylor estimate.

With $a = 5\rho^2$, contraction by $s = (1 + 2a)^{-1/2}$ repairs every point of the relaxation into C . The Euclidean displacement is at most $\sqrt{2}a$. The N equally spaced angles have covering radius $\pi/(4(N-1))$, giving the displayed upper constant. The estimate concerns the entire relaxation, not only selected witnesses.

A finite-grid lower bound. The verified lower proof uses $N+1$ parameters $\tau_j = 1 + j/N$, $j = 0, \dots, N$, and the rational perturbation $\eta = 1/(400N^2)$. It realizes the Gram matrices

$$\begin{pmatrix} 1 - 1/(10\tau_j) & 2/5 - \eta \\ 2/5 - \eta & 1 - \tau_j/10 \end{pmatrix}$$

by actual vectors in every required dimension. If one good multiplier excludes witnesses at two parameters $\tau, \sigma \in [1, 2]$, the cone inequality gives

$$(\tau - \sigma)^2 < 16((1 + 10\eta)^2 - 1) < 1/N^2.$$

Thus a cut excludes at most one grid point, and at most N cuts leave one witness admitted. This finite pigeonhole argument needs no interval-covering theorem; it includes zero-coordinate and interior multipliers and imposes no bound on their ratios.

At an admitted witness, the good multiplier $(\tau, 1/\tau, 2)$ has value 2η . Its quadratic has Lipschitz constant at most 10 on the product of the two Euclidean unit balls, where both the witness and C lie. Every hull point therefore has distance at least $\eta/5$ from the witness. This proves the stronger lower estimate

$$d_H(C, P_F) \geq \frac{1}{2000N^2} \geq \frac{\sqrt{2}(\log 2)^2}{1600N^2}.$$

The final comparison is also proved in Lean. The main paper improves the first lower constant to $\sqrt{2}/2000$ using a sharper Lipschitz estimate; that improvement is not part of this formal package.

Exact rational coefficients. For $m \geq 1$, use

$$(m^2, j^2, 2mj), \quad (j^2, m^2, 2mj), \quad 0 \leq j \leq m.$$

The common middle ray is counted once, leaving exactly $2m+1$ distinct cuts. They are good and include both coordinate rays. Their angles have covering radius at most $1/(2m)$, proved using the Lipschitz bound for arctangent. Positive rescaling connects these exact integer coefficients to the angle-family theorem, yielding error at most $5\sqrt{2}/(4m^2)$.

For $N \geq 3$, choosing $m = \lfloor (N-1)/2 \rfloor$ meets the cut budget. Each coefficient is a nonnegative integer at most $2m^2$. With the binary size convention assigning zero size to zero, the package bounds each size by $2 \text{size}(N) + 1$, and hence by $2 \lfloor \log_2 N \rfloor + 3$. It also proves a uniform constant times N^{-2} error bound for these rational families. These statements concern exact representation; they do not analyze numerical rounding or solver conditioning.

Rates, tolerances, and scope. The extended-real family bounds imply finiteness of e_N for $N \geq 2$ before it is converted to a real number. The package then proves the asymptotic $\Theta(N^{-2})$ statement. Writing $c = \sqrt{2}(\log 2)^2/1600$ and $B = 5\sqrt{2}\pi^2/16$, any family of at most $N \geq 1$ cuts with error at most $\varepsilon > 0$ must satisfy $N \geq \sqrt{c/\varepsilon}$. Conversely the explicit angle family with at most $\lceil \sqrt{B/\varepsilon} \rceil + 1$ cuts suffices; its size is less than $\sqrt{B/\varepsilon} + 2$. Sufficiency constructs a family and does not infer attainment from an infimum estimate.

The frozen claims and declaration coverage are in `supplement/lean/audits/31-aggregation-accuracy/`. The support-function interpretation and single-objective exact-aggregation proposition in the main paper are outside this package. These uniform approximation bounds give no solver runtime, iteration, or branch-and-bound lower bound, and make no literature-priority claim.

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